

Docker Volumes & Storage

Volumes · Bind Mounts · tmpfs · Data Persistence

■ Analogy

- Container storage is like a hotel room. The room itself (container layer) is temporary — cleaned when you check out. A USB drive you plug in (bind mount) brings your own data. The hotel safe (Docker volume) persists even after you leave and can be moved between rooms.

Key Terms

Volume	Docker-managed storage on the host. Lives in <code>/var/lib/docker/volumes</code> . Survives container removal
Bind Mount	Mounts a host directory into the container. Good for dev — changes reflect instantly
tmpfs mount	In-memory storage. Fast, but lost when container stops. Good for secrets/temp data
Copy-on-Write	Container's writable layer — modifications don't touch the base image layers
Named Volume	<code>docker volume create mydata</code> — reusable, manageable with <code>docker volume</code> commands
Anonymous Volume	Created without a name — hard to manage, avoid in production
-v flag	Legacy: <code>-v mydata:/app/data</code> — volume mount. Also <code>-v /host/path:/container/path</code> for bind
--mount flag	Explicit, verbose syntax: <code>--mount type=volume,source=mydata,target=/data</code>
Volume Driver	Plugin to store volumes in cloud storage — AWS EBS, NFS, Flocker, etc.
Overlay2	Docker's default storage driver on Linux — uses OverlayFS for image layers

Storage Types Comparison

Type	Persists?	Managed by Docker?	Use Case
Volume	Yes	Yes	Databases, persistent app data
Bind Mount	Yes (host file)	No	Dev: live code reload, config files
tmpfs	No (memory only)	Yes	Secrets, temp files, cache
Container layer	No (removed with container)	Yes	Ephemeral scratch space

Volume Commands & Examples

```
# Create a named volume
docker volume create pgdata

# Run Postgres with persistent volume
docker run -d \
  --name postgres \
  -v pgdata:/var/lib/postgresql/data \
  -e POSTGRES_PASSWORD=secret \
  postgres:15

# Bind mount for live code reloading (dev only!)
docker run -d \
  --name myapp \
  -v $(pwd)/src:/app/src \
  myapp:dev

# tmpfs for sensitive in-memory data
docker run -d --mount type=tmpfs,target=/run/secrets myapp

# Inspect volume location on host
docker volume inspect pgdata

# Back up a volume
docker run --rm -v pgdata:/data -v $(pwd):/backup \
  alpine tar czf /backup/pgdata.tar.gz /data

# Remove unused volumes
docker volume prune
```

■ Real-World: GitLab

- GitLab uses Docker volumes for persistent storage of PostgreSQL, Redis, and Git repository data in their self-hosted Docker deployment. Named volumes ensure data survives container upgrades — GitLab ships `docker-compose.yml` with explicit named volumes so operators can upgrade by replacing containers without losing repository data.